

Course Code	Name of the Course				
216U03C503	Digital Signal Processing				
Teaching Scheme	TH	P	TUT	Total	
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Credits Assigned	03		--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		20	30	50	100

Course pre-requisites:

- Signals and systems
- Mathematics for Communication Engineering

Course Objectives:

Signal processing is an area of science and engineering developed over decades. This fundamental course addresses the representation, analysis, and design of discrete time Systems. The major concepts covered include Discrete-time processing of continuous-time signals in time and frequency domain, discrete Fourier transform, FFT algorithm, structures for DT systems, design techniques for recursive (IIR) and non-recursive (FIR) filters; multirate techniques.

Course Outcomes (CO):

Course Outcomes

At the end of successful completion of the course the student will be able to

- CO1.** Analyze discrete time systems in time and Z-domain
- CO2.** Realize discrete system using different structures
- CO3.** Analyze discrete time systems using discrete Fourier transform
- CO4.** Design Finite Impulse Response and Infinite Impulse Response filters
- CO5.** Understand concept of Multi-rate signal processing

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Introduction: Discrete Time (DT) systems and analysis in time and Z-domain		10	CO1 CO2
	1.1	Review of discrete time signals and systems: Concept of auto-correlation, cross-correlation, Linear and circular convolution, Impulse response, transfer function.		
	1.2	LTI system as Frequency selective filters: Ideal filter characteristics, Low pass, High pass, Band pass, band stop, Notch, resonator, averaging, comb, all pass filters. Pole Zero placement and analysis, Structures for realization of systems: Direct form I and II, cascade, parallel.		
	1.3	Introduction to DSP systems and their applications in areas such as Biomedical and Communication systems		
2	Analysis of discrete time system in frequency domain		10	CO3
	2.1	Discrete Time Fourier Transform (DTFT) : analysis of Linear Time Invariant (LTI) DT systems using DTFT.		
	2.2	Discrete Fourier Transform (DFT): Frequency domain sampling, Properties of DFT, Filtering of long Data sequences, Analysis of LTI DT system using DFT, aliasing in frequency domain.		
	2.3	Fast Fourier Transform (FFT): Efficient computation of DFT, Radix 2 Decimation In Time (DIT) and Decimation In Frequency (DIF) FFT algorithms for 8-point DFT computation.		
3	Design of FIR Filters		09	CO4
	3.1	Characteristics of FIR filters with linear phase, symmetric and antisymmetric sequences.		
	3.2	Design specification of digital filters.- FIR filter design using windowing techniques : Design with rectangular, Hamming, and Hanning window		
	3.3	FIR filter design using frequency sampling method.		
4	Design of IIR Filters		10	CO4
	4.1	Frequency response of analog and digital IIR filter, Analog Butterworth and Chebyshev-1 filters (No derivation for transfer function, Frequency transformations- Analog, digital		
	4.2	Low pass digital IIR filter design using Impulse Invariance and Bilinear Transformation method using Butterworth and Chebyshev-1 approximation		

5	Multi rate Digital Signal Processing		06	CO5
	5.1	Decimation and Interpolation by integer factor		
	5.2	Sampling rate conversion, Poly phase decomposition		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1.	Sanjit Mitra	<i>Digital Signal Processing- A computer Based Approach</i>	TATA McGraw Hill, India	4th Edition, 2013
2.	A.V. Oppenheim	<i>Digital Signal Processing</i>	Prentice Hall, India	2nd Edition, 2013
3.	John Proakis, Dimitris Manolakis	<i>Digital Signal Processing, Principles, Algorithms and Applications</i>	Prentice Hall, India	4th Edition, 2005
4.	A. Nagoor Kani	Digital Signal Processing	TATA McGraw Hill, India	2nd Edition, 2013
5.	Ashok Ambardar	<i>Digital Signal Processing: A Modern Introduction</i>	CENGAGE Learning India	1st Edition, 2004